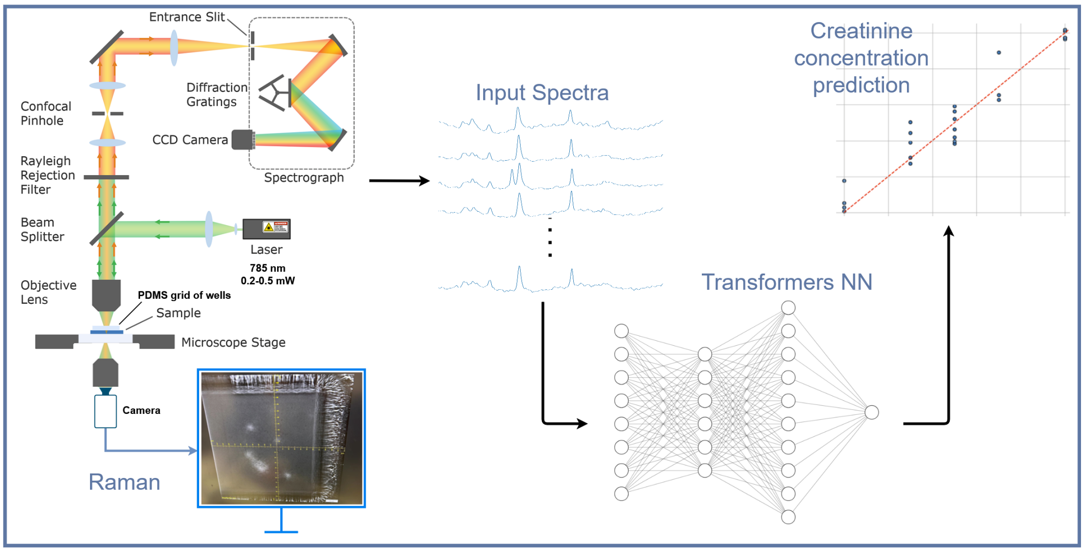
Surface-Enhanced Raman Spectroscopy for Quantitative, Label-Free Creatinine Sensing Using Transformers Neural Networks



Daniel Abraham Elmaleh

Semester Project
Master's in Microengineering

EPFL

Professor: Hatice Altug
Supervisor: Prof Hatice Altug

Location: Bio-Nano-Photonic Systems laboratory
Date: June 26, 2025

Abstract

Continuous, label-free monitoring of creatinine is essential for early diagnosis and management of kidney disease. This project investigates the feasibility of using surface-enhanced Raman spectroscopy (SERS) combined with machine learning to achieve quantitative detection of creatinine in phosphate-buffered saline (PBS). SERS measurements were performed on SERS substrates optimized for 785 nm excitation, with samples spanning concentrations from 30 to 300 µM. A confocal Raman microscope was used to collect spectra from dried micro-spotted samples. The processed data served as an input yo a transformer-based neural network architecture. The model was developed to predict creatinine concentrations from Raman spectra using vector regression (SVR) on embeddings from the penultimate layer of the NN model. Yielding strong predictive performance ($RMSE < 20\,M, R > 0.95$ for the full range; $RMSE < 10\,M, R > 0.88$ for 0–100 M). These results demonstrate the potential of combining SERS with advanced machine learning for sensitive, quantitative detection of creatinine in PBS. Future work will focus on improving reproducibility, extending to serum samples, and developing an automated, portable sensing device with integrated microfluidics for continuous monitoring.